

COURSE NO.:	IPE 6112		
COURSE TITLE:	Probabilistic Methods in Engineering Design		
INSTRUCTOR:	Kais Zaman, PhD Dept. of Industrial and Production Engineering Email: kaiszaman@ipe.buet.ac.bd		
TEXT :	"Probability, Reliability and Statistical Methods in Engineering Design," A. Haldar and S. Mahadevan, John Wiley & Sons, 2000.		
SCHEDULE:	Wednesday 3:00 to 6:00 p.m., IPE 304.		
GRADING:	Homework	20%	
	2 mid-term exams	30%	
	Term Project	20%	
	Final exam	30%	
EXAM SCHEDULE:	Mid-term Exam 1	Wednesday, TBA	
	Mid-term Exam 2	Wednesday, TBA	
	Term Project Due	Wednesday, TBA	
	Final Exam	Wednesday, TBA	

COURSE DESCRIPTION:

Course Overview:

The planning and design of most engineering systems are inherently conducted under conditions of uncertainty. Many system parameters are not precisely known, and subjective decisions are required throughout the design process. Traditionally, uncertainties have been addressed by employing empirical safety factors. However, in recent decades, this approach has been progressively replaced by more rational methodologies that quantify uncertainties and develop design rules based on acceptable levels of failure probability. As a result, knowledge of probability and statistics has become crucial in the design of modern engineering systems.

This course aims to equip students with the essential techniques for applying probability and statistics to a wide range of engineering problems. The course requires no prior knowledge of probability theory, and the emphasis will be on practical applications in various fields, including industrial, financial, civil, mechanical, chemical, and electrical engineering.

Course Objectives:

- Understand and quantify uncertainties in engineering systems.
- Apply statistical methods to model and analyze data.
- Utilize probabilistic approaches for reliability analysis and design.

Term Project:

Students will complete a term project that may take one of three forms:

1. **Research on Advanced Topics:** Investigate advanced topics related to probabilistic methods in engineering.
2. **Literature Review and Case Studies:** Review relevant literature and provide examples of applications in engineering.
3. **Practical Application:** Apply class concepts to solve a practical engineering problem.

The project will include both a written report and an oral presentation.

Topics Covered:

1. **Basic Probability Concepts:** Introduction to probability, events, and outcomes.
2. **Random Variables and Distributions:** Understanding discrete and continuous random variables and their probability distributions.
3. **Data Analysis:** Techniques for identifying distributions and estimating parameters from data.
4. **Functions of Random Variables:** Analysis of the behavior of functions involving random variables.
5. **Response Surface Methods:** Approximating complex models and exploring the impact of uncertainty.
6. **Monte Carlo Simulation:** Using random sampling methods for uncertainty propagation and probabilistic analysis.
7. **Component Reliability Analysis:** First-order and second-order methods for assessing component reliability.
8. **System Reliability:** Evaluating the reliability of systems composed of multiple components.
9. **Reliability-Based Design:** Designing engineering systems to meet specified reliability targets.